University of Asia Pacific

Department of Computer Science and Engineering

CSE 438: Robotics Lab

LAB REPORT

Experiment Number: 05

Experiment Title: Bluetooth Based Android Controlled Robot

Submitted by:

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1. Experiment Name

Bluetooth Based Android Controlled Robot

2. Objective

The main objectives of the Bluetooth Based Android Controlled Robot project are:

- To design and develop a robot that can be controlled wirelessly using an Android smartphone.
- To establish Bluetooth communication between the Android device and the microcontroller.
- To control the robot's movement in different directions (forward, backward, left, right, stop).
- To interface DC motors with a microcontroller using a motor driver.
- To indicate robot movement directions using LED indicators.

3. Apparatus / Hardware & Software Requirements

Hardware Requirements

- Microcontroller (PIC16F877A / AT89C51)
- Bluetooth Module (HC-05 / HC-06)
- DC Motors (2 units)
- Motor Driver IC (L293D / L298)
- Crystal Oscillator
- Reset Circuit
- LED Indicators
- 16×2 LCD (optional)
- Regulated Power Supply (7805 Voltage Regulator)
- Battery (9V or 12V)
- Connecting wires
- Robot chassis
- Wheels

Software Requirements

- Thinkercad

4. Circuit Diagram / Schematic

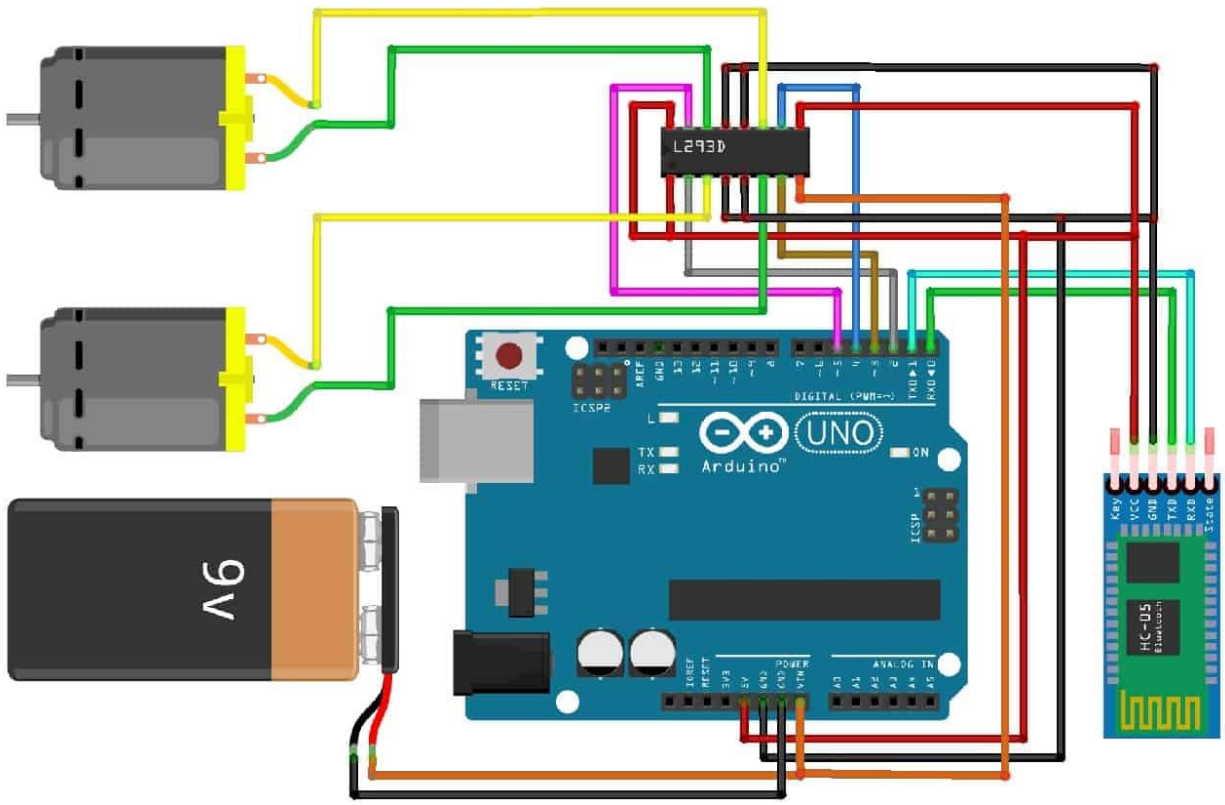

Fig:1 - Design of a Self-Balancing Robot

5. Code / Program

```
#include <xc.h>

#define _XTAL_FREQ 20000000

void UART_Init() {

    TXSTA = 0x20;    // Transmit Enable

    RCSTA = 0x90;    // Serial Port Enable

    SPBRG = 129;     // Baud Rate 9600
```

```
}

char UART_Read() {
    while(!RCIF);
    return RCREG;
}

void main() {

    TRISB = 0x00;    // PORTB as output
    PORTB = 0x00;

    UART_Init();

    char data;

    while(1) {

        data = UART_Read();

        if(data == 'F') {           // Forward
            PORTB = 0x05;
        }

        else if(data == 'B') {    // Backward
            PORTB = 0x0A;
        }

        else if(data == 'L') {    // Left
            PORTB = 0x06;
        }
```

```
        else if(data == 'R') {     // Right
            PORTB = 0x09;
        }

        else if(data == 'S') {     // Stop
            PORTB = 0x00;
        }
    }
}
```

6. Output / Observations

After successful implementation, the following outputs were observed:

1. Bluetooth connection between smartphone and robot was established successfully.
2. Commands sent from Android app were received by the microcontroller.
3. Robot responded according to commands:

Command	Robot Action
F	Move Forward
B	Move Backward
L	Move Left
R	Move Right
S	Stop

Table:1 - Robot responded

4. LED indicators displayed robot movement direction.
5. Motors operated smoothly based on received commands.

6. Result

The Bluetooth-based Android controlled robot system was successfully designed and implemented.

The robot responded correctly to commands transmitted from the Android smartphone via Bluetooth. The microcontroller processed the received signals and controlled the DC motors accordingly, allowing movement in multiple directions.

Wireless communication was achieved effectively using Bluetooth technology.

8. Conclusion

The Bluetooth Based Android Controlled Robot was successfully developed using a microcontroller and Bluetooth communication system. The project demonstrated reliable wireless control of robot movement using an Android smartphone.

This system reduces human effort by enabling remote operation and can be applied in surveillance, industrial automation, and rescue operations. Future improvements may include adding sensors, cameras, and advanced control features to enhance the robot’s performance and functionality.